



Subject: RFP — CIF Quotation Request — 12 MW / 48 MWh BESS — South Africa (Standalone Order)

To: Kamil Talar (kamil@linyang.com)

Priority: HIGH

Response by: 24 March 2026

Dear Kamil,

I hope this finds you well.

We have a new standalone BESS project opportunity in **South Africa** and would like to request a CIF quotation from Linyang. This is **not part of the Cyprus group order** — it is an independent project with a separate client and delivery destination.

The project has received its **Eskom Planning Report** (grid connection officially secured), and we need indicative pricing urgently to prepare our EPC proposal.

1 Project Summary

Parameter	Value
Project Name	DMSH Geelvloer Facility
Location	Kenhardt area, Northern Cape, South Africa
Client	AE Alternative Energy GmbH (Germany)
EPC Contractor	Lighthief Ltd
Grid Operator	Eskom
Project Type	Private IPP — PPA with Eskom
Application	Utility-scale 75 MWp solar PV + BESS hybrid
BESS Purpose	Grid stability, energy shifting, dispatchability

2 BESS Technical Requirements (Eskom Final — March 2026)

Parameter	Value
BESS Power	12 MW (fixed, Eskom confirmed)
BESS Energy	48 MWh (4-hour duration)

C-Rate	0.25C charge / 0.25C discharge
Architecture	AC-coupled
Internal MV Voltage	33 kV
POC Voltage	132 kV (Eskom transmission)
Frequency	50 Hz
System Fault Level	4.55 kA at 132 kV — all switchgear/protection must be rated accordingly
Grid Code	South African BESF Code v5.3 — Category C compliance mandated by Eskom

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Grid Code Compliance Requirements (SA BESF v5.3 + RPP Rev 3.1)

Eskom mandates full Category C compliance. Key requirements:

Requirement	Value
LVRT	0 pu (zero voltage) for 150 ms — must stay connected
HVRT	120% voltage for 2 seconds
RoCoF withstand	±2.5 Hz/s (200 ms measurement window)
Reactive power	±0.33 × Pn = ±3.96 MVar (PF 0.95)
Frequency range	47.0 – 52.0 Hz
Continuous frequency	49.0 – 51.0 Hz
Deadband	49.85 – 50.15 Hz
Droop (mandatory FSM)	4%
Droop (LFSM-U / LFSM-O)	10%
Active power recovery	≥90% within 1 second post-fault
Reactive current injection	Within ±20% tolerance after 60 ms
Phase jump tolerance	20°
Anti-islanding	Detection + disconnect within 2 seconds
Ramp rate	1% – 20% of Pn per minute (adjustable)
SCADA protocol	IEC 60870-5-101 (transmission connection)
PQ monitoring	IEC 61000-4-30 Class A

Question for Linyang: Can you confirm that the Linyang BESS + Kehua PCS platform meets all of the above SA grid code requirements? If any item requires additional equipment or certification, please specify.

4 Environmental & Site Conditions

Parameter	Value	Design Impact
Ambient Temperature Max	45°C	PCS and HVAC derating required
Ambient Temperature Min	-5°C	Cold start / heating required
Altitude	916 m	Air density reduction — cooling derating (~5-10%)
Pollution Level	Heavy (semi-desert, dust)	IP-rated switchgear, sealed cabinets, dust filters
Terrain Slope	Max 9.2% N-S	Foundation design consideration
Distance from Coast	≥10 km (≈400 km inland)	Standard corrosion protection adequate

Questions for Linyang:

1. What is the PCS power derating at 45°C ambient and 916 m altitude? Please confirm continuous output at these conditions.
2. What IP rating is provided for the PCS skid and battery containers? Is heavy pollution / dust environment covered by standard design?
3. Are additional HVAC upgrades or dust filtration options required for this environment?

5 Quotation Request — Two Scenarios

We request CIF pricing for **two scenarios** to allow the client to choose:

Scenario A: Grid Following Only

Item	Specification
Battery containers	12 MW / 48 MWh (standard Linyang LFP containers)
PCS / Inverters	Grid Following — standard Kehua PCS
MV Transformer	0.63/33 kV, Dyn11, ONAN
MV Switchgear	Rated ≥ 4.55 kA
Containerised PCS skid(s)	Per Linyang standard configuration

Scenario B: Grid Forming (150% Overload for 10 seconds)

Item	Specification
Battery containers	12 MW / 48 MWh (standard Linyang LFP containers)
PCS / Inverters	Grid Forming — 150% overload (18 MW) for 10 seconds

MV Transformer	0.63/33 kV, Dyn11, ONAN — sized for overload
MV Switchgear	Rated ≥ 4.55 kA
Containerised PCS skid(s)	Per Linyang configuration for grid forming

For each scenario, please provide:

1. **CIF price (Cape Town, South Africa)** — itemised:
 - Battery containers (total)
 - PCS / inverters (total)
 - MV transformer
 - MV switchgear
 - Containerised PCS skid(s)
 - Cabling / accessories
 - Freight (Shanghai → Cape Town)
2. **Container count** and individual container capacity (MWh)
3. **PCS model and quantity** (which Kehua model?)
4. **Skid configuration** (T1/T2/T4/T8?)
5. **Total shipping weight and dimensions** (for transport planning)
6. **Production lead time** from order confirmation
7. **Shipping time** (Shanghai → Cape Town)

6 Delivery & Timeline

Parameter	Value
Incoterms	CIF Cape Town, South Africa
Target Delivery	August 2026 (Q3)
Port of Discharge	Cape Town

Question for Linyang: Given the August 2026 target, what is the latest date a production order must be placed? Please confirm production lead time (90 days per Cyprus standard?) and shipping duration to Cape Town.

7 Warranty & Support

Parameter	Request
Base Warranty	5 years (per Linyang standard)
Extended Warranty	Please quote Yr 6-10 and Yr 11-15 rates (per MWh, consistent with Cyprus portfolio pricing)

SOH Guarantees	Please confirm SOH curve applies (85% @ Yr 5, ~80% @ Yr 10, 70% @ Yr 15)
Cycle Life	7,000 cycles at 70% EOL DoD

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Scope Clarification

Included in Linyang CIF scope (please confirm):

- Battery containers (fully assembled, tested)
- PCS / storage inverters
- MV transformer (LV/MV step-up)
- MV switchgear
- Containerised PCS skid(s)
- Internal cabling and accessories
- Marine freight (Shanghai → Cape Town)
- Marine cargo insurance

NOT included (Lighthief scope):

- HV transformer (33/132 kV) — TBC
- 132 kV switchgear and interconnection
- Civil works, foundations, fencing
- EMS / SCADA (Lighthief / Voltus)
- IEC 60870-5-101 gateway
- IEC 61000-4-30 Class A PQ monitoring
- Commissioning and grid code compliance testing
- Local installation labour in South Africa

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Commercial Terms

Parameter	Request
Payment Terms	Please propose (reference: Cyprus portfolio is 20/50/20/10)
Quotation Validity	Minimum 60 days
Currency	EUR or USD — please specify
Performance Bond	Required? Same terms as Cyprus (5% bank guarantee)?

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Documentation Required with Quotation

Please provide with your quotation:

1. Itemised CIF pricing (both scenarios)

2. PCS datasheet for the proposed model (with derating curves for 45°C / 916 m)
 3. Battery container datasheet (capacity, dimensions, weight, IP rating)
 4. Preliminary GA drawing / container layout for 12 MW / 48 MWh
 5. SA grid code compliance confirmation or gap analysis
 6. Production and shipping timeline
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Response requested by: 24 March 2026

This is a time-sensitive opportunity — the client has received Eskom grid connection approval and needs our EPC proposal promptly.

Thank you, Kamil. Please let me know if you need any additional information.

Best regards,



Alexander Papacosta

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